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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

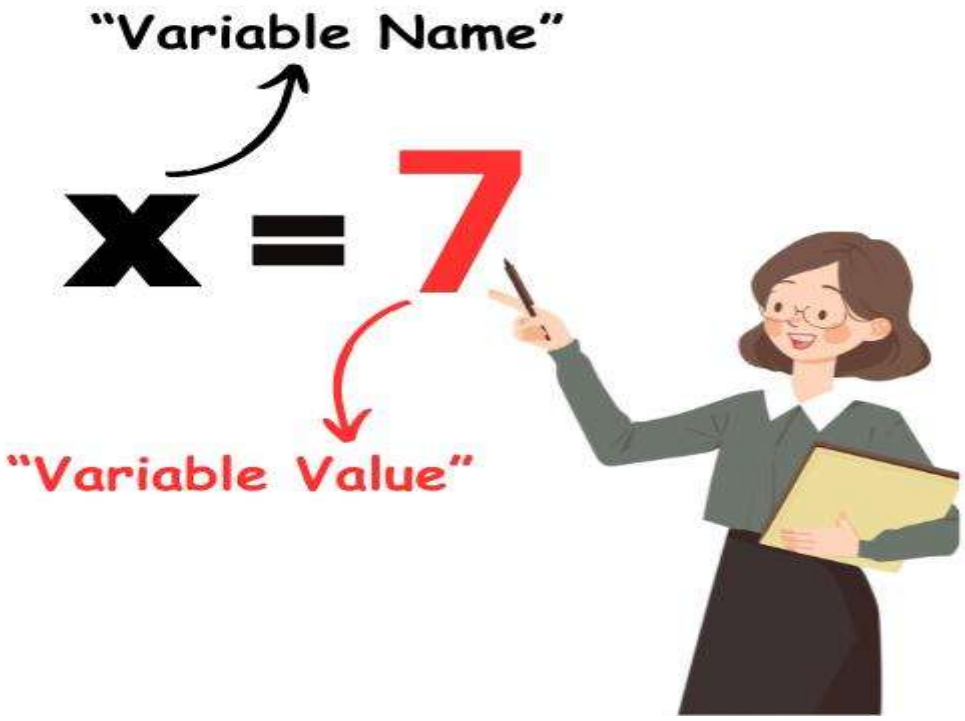
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 15: Variables



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Learning Objectives

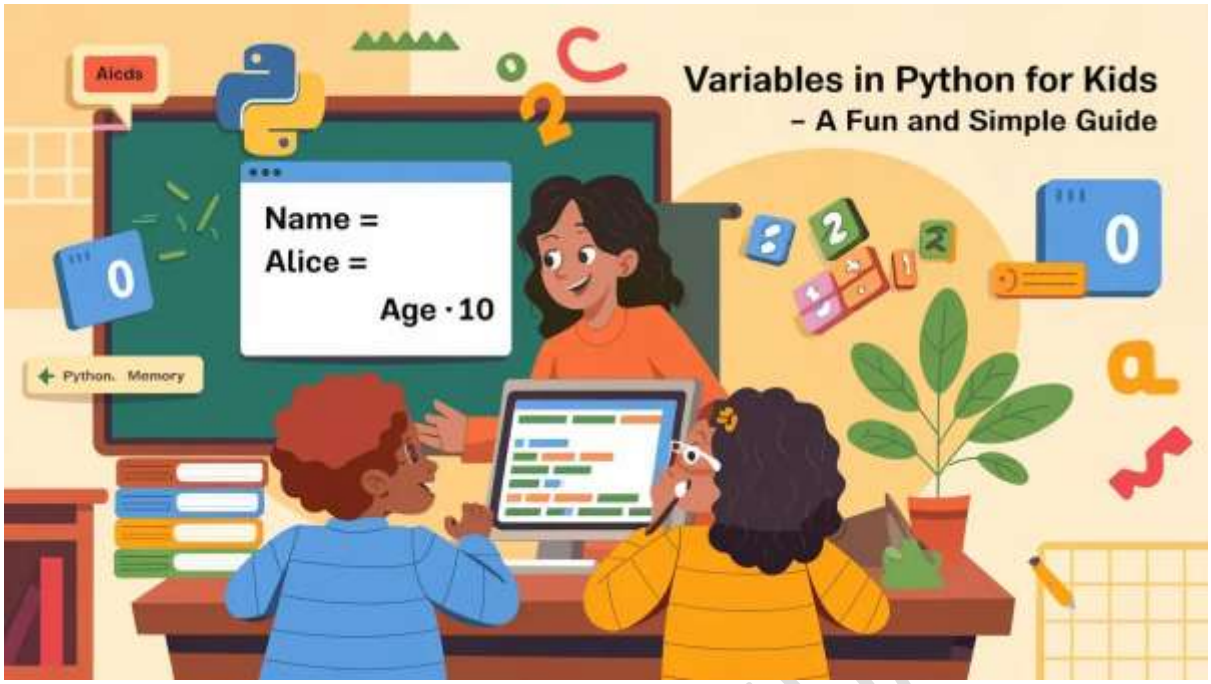
By the end of this lesson, students will be able to:

- Understand what a variable is.
- Learn why variables are useful.
- Create variables in Python.
- Store text, numbers, and decimal values.
- Print variable values.
- Follow simple variable naming rules.
- Create small Python programs using variables.

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What is a Variable?





A **variable** is like a **box** or **container** that stores information.

Imagine you have different boxes:

📦 Box 1 → Your Name

📦 Box 2 → Your Age

📦 Box 3 → Your Favorite Color

In Python, variables store information that we can use later.

Real-Life Example

Imagine these school items:

🎒 School Bag → Stores books

✏️ Pencil Box → Stores pencils

🏦 Bank Account → Stores money

📦 Variable → Stores data

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A variable keeps information safely until we need it.

Why Do We Use Variables?

Variables help us:

- Store information
- Reuse information
- Change information easily
- Make programs shorter and smarter

Instead of typing the same information many times, we store it once in a variable.

Creating a Variable

In Python, we use the **equal sign (=)** to store a value.

Example:

name = "Rahul"

Here:

- name → Variable name
- = → Assignment operator
- "Rahul" → Value

Printing a Variable

name = "Rahul"

print(name)

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Output

Rahul

Notice: **Do not use quotation marks when printing a variable.**

Example: Student Name

```
student = "Ananya"


print(student)
```

Output

Ananya

Variables with Numbers

```
age = 12


print(age)
```

Output

12

Numbers do **not** need quotation marks.

Decimal Numbers

Python can also store decimal numbers.

```
height = 4.8


print(height)
```

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Output

4.8

Favorite Subject

```
subject = "Science"


print(subject)
```

Output

Science

Multiple Variables

```
name = "Aarav"
age = 13
school = "ABC High School"


print(name)
print(age)
print(school)
```

Output

Aarav

13

ABC High School

Different Types of Variables

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Variable	Example	Type
Name	"Riya"	Text (String)
Age	13	Integer
Height	4.9	Decimal (Float)
Is Student	True	Boolean

Displaying a Sentence

```
name = "Rohan"


print("Student Name:", name)
```

Output

```
Student Name: Rohan
```

Classroom Example

```
student = "Neha"
class_name = 7
favorite_game = "Chess"


print(student)
print(class_name)
print(favorite_game)
```

Output

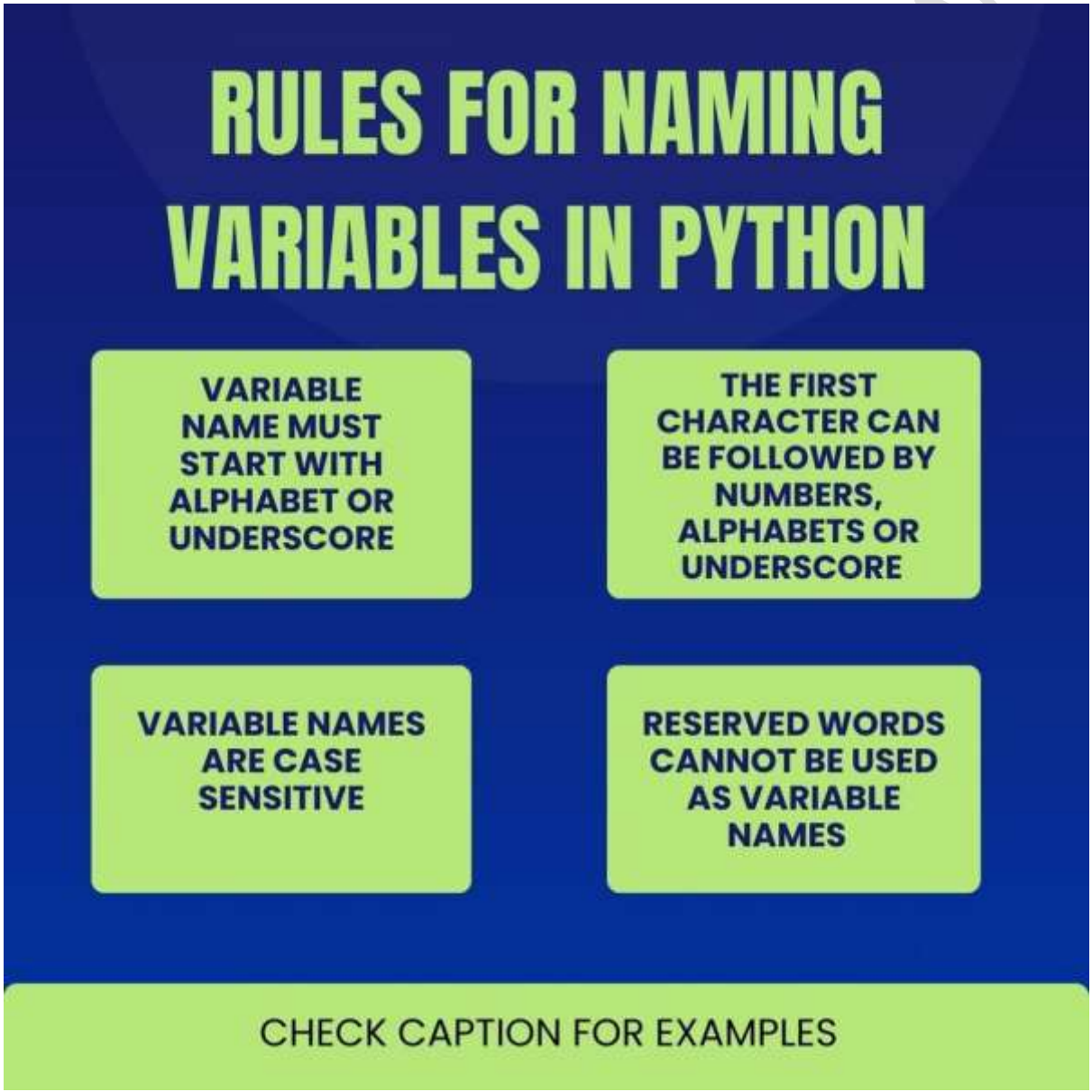
```
Neha
```

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Chess

Variable Naming Rules



CODESEMPAI

VARIABLES-NAMES

A variable can have a short name (like x and y) or a more descriptive name (age, carname, total_volume).

Legal variable names:

```
myvar = "Naruto,"
my_var = "hinata,"
_my_var = "sasuke,"
myVar = "Juraya,"
MYVAR = "kakashi,"
myvar2 = "Akatsuki"

print(myvar,my_var,_my_var, myVar,MYVAR,myvar2)
```

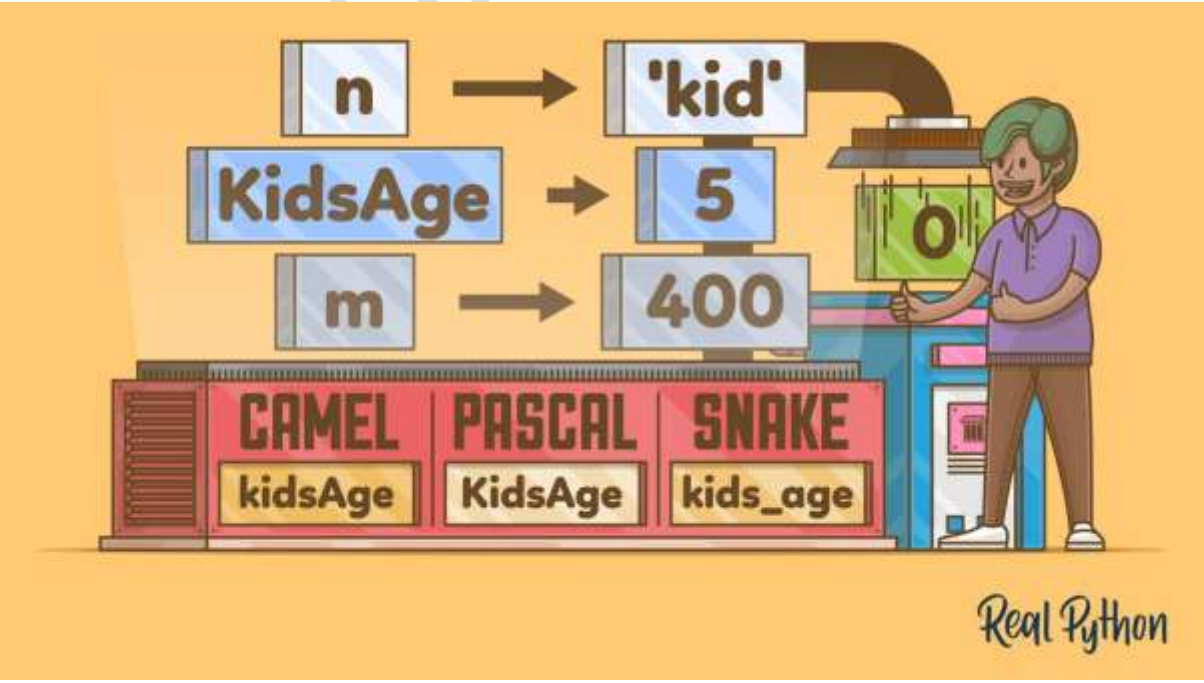
```
Naruto, hinata, sasuke, Juraya, kakashi, Akatsuki
```

Illegal variable names:

```
2myvar = "Naruto,"
my var = "hinata,"
my var = "sasuke,"
print(myvar,my_var,_my_var)
```

```
File "<string>", line 1
    2myvar = "Naruto,"
    ^
SyntaxError: invalid syntax
```

Remember that variable names are case-sensitive



✅ Variable names can contain:

- Letters
- Numbers
- Underscore (_)

Examples:

student_name

age

marks1

favorite_color

❌ Wrong Variable Names

1name

my-name

student name

class@

Good vs Bad Variable Names

Good	Bad
student_name	student name
total_marks	total-marks
age	123age
school_name	school#

Changing a Variable

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Variables can change.

```
score = 10


print(score)


score = 20


print(score)
```

Output

10

20

Using Variables in Math

```
a = 5

b = 3


print(a + b)
```

Output

8

Fun Activity

```
favorite_color = "Blue"

favorite_food = "Pizza"

favorite_animal = "Tiger"
```

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```
print(favorite_color)

print(favorite_food)

print(favorite_animal)
```

Practical Activities

Practical Activity

Students should create a file named:

variables.py

Write the following program:

```
name = "Your Name"

age = 12

school = "Your School"


print(name)

print(age)

print(school)
```

Replace the values with your own information.

Classroom Activity

Ask students to create variables for:

- Name
- Age

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- Class
- Favorite Subject
- Favorite Sport

Example:

```
name = "Priya"
age = 13
class_name = 8
subject = "Math"
sport = "Badminton"


print(name)
print(age)
print(class_name)
print(subject)
print(sport)
```

Mini Challenge

Create a file named:

myprofile.py

Write a program that stores and prints:

- Name
- School
- City
- Favorite Book

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- Dream Career

Example:

```
name = "Rahul"
school = "Sunrise School"
city = "Hyderabad"
book = "Panchatantra"
career = "Software Engineer"

print("Name:", name)
print("School:", school)
print("City:", city)
print("Favorite Book:", book)
print("Dream Career:", career)
```

Common Mistakes

❌ **Using Spaces**

Wrong:

```
student name = "Riya"
```

Correct:

```
student_name = "Riya"
```

❌ **Starting with a Number**

Wrong:

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1age = 12

Correct:

age1 = 12

❌ **Printing Text Without Quotes**

Wrong:

print(Hello)

Correct:

print("Hello")

Important Terms

Word	Meaning
Variable	Stores data
Value	Information stored in a variable
Assignment (=)	Stores a value in a variable
String	Text data
Integer	Whole number
Float	Decimal number
Boolean	True or False

Worksheet

Fill in the Blanks

1. A _____ stores information in Python.

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Answer: Variable

2. The assignment operator is _____.

Answer: =

3. Text values are written inside _____.

Answer: Quotation marks

4. Whole numbers are called _____.

Answer: Integers

5. Decimal numbers are called _____.

Answer: Floats

Match the Following

A	**B**
Variable	Stores data
String	Text
Integer	Whole number
Float	Decimal number
=	Assignment operator

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Quiz

1. What is a variable?

A. A game

B. A storage box for data ✅

C. A computer

D. A browser

2. Which operator is used to assign a value?

A. +

B. = ✅

C. *

D. /

3. Which is a valid variable name?

A. student name

B. 1student

C. student_name ✅

D. student-name

4. What is the output?

age = 12

print(age)

A. age

B. "age"

C. 12 ✅

D. Error

5. Which data type stores text?

A. Integer

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B. Float

C. String ✅

D. Boolean

Learning Outcomes

After completing this lesson, students will be able to:

- ✅ Explain what a variable is using real-life examples.
- ✅ Create variables to store text and numbers.
- ✅ Print variable values correctly.
- ✅ Follow Python variable naming rules.
- ✅ Update variable values when needed.
- ✅ Write simple Python programs using variables confidently.

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